

Snake Game

Clone Repo : <https://github.com/codertheashish/Snake-Game>

1. Introduction

The **Snake Game** is a classic arcade game developed using **Python Turtle Graphics**. In this game, the player controls a snake using the keyboard arrow keys. The snake grows longer after eating food, and the score increases. The game ends when the snake hits the wall or its own body. The high score is saved until the program is closed.

2. Purpose

The main purpose of this project is to learn the basics of **Python game development** and improve programming skills.

This project helps in understanding:

- Turtle Graphics
 - Keyboard events
 - Loops
 - Functions
 - Collision detection
 - Random object generation
 - Score management
 - Real-time game logic
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3. Technologies Used

Technology	Purpose
Python	Main programming language
Turtle Module	Draws the game objects and window
Random Module	Generates random food position, shape, and color
Time Module	Controls the game speed

4. Main Features

- Five speed modes
 - Keyboard controls
 - Random food generation
 - Snake body growth
 - Score and high score system
 - Wall collision detection
 - Self-collision detection
 - ESC key to exit the game
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5. Project Workflow

```
Start Game
|
Choose Speed
|
Create Window
|
Create Snake
|
Create Food
|
Wait for Keyboard Input
|
Move Snake
|
Check Food Collision
|
Increase Score
|
Grow Snake
|
Check Wall Collision
|
Check Self Collision
|
Game Reset (if collision occurs)
```
